M3503 WEEK 3 2021

HIGHER ORDER LINEAR DIFFERENTIAL EQUATIONS

HOMOGENEOUS LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

An nth order differential equation of the form

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = f(x)$$

is homogeneous if $f(x) = 0$ and non-homogeneous if $f(x) \neq 0$

SUPERPOSITION PRINCIPLE FOR HOMOGENEOUS LINEAR EQUATIONS

If $y_1, y_2, \ldots y_k$ are solutions to a homogeneous linear equation, then $y = c_1 y_1 + c_2 y_2 + \cdots + c_k y_k$, where c_i's are constants, is also a solution to the homogeneous linear equation.

LINEAR DEPENDENCE/INDEPENDENCE

A set of functions $y_1, y_2, \ldots, y_n$ is linearly dependent on an interval I if there exist constants $c_1, c_2, \ldots, c_n$, not all zeros, such that $c_1 y_1 + c_2 y_2 + \cdots + c_n y_n = 0$ on I, otherwise, it is independent.

For solutions of differential equations, we are mostly interested in linearly independent solutions, also known as a fundamental set of solutions.

Suppose each of the solutions $y_1, y_2, \ldots, y_n$ has at least $n-1$ derivatives, the determinant

$$W(y_1, y_2, \ldots, y_n) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & & y_n' \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}$$

is called the **Wronskian** of the functions $y_1, y_2, \ldots, y_n$

Criterion For Linearly Independent Solutions

Let $y_1, y_2, \ldots, y_n$ be solutions to an nth order homogeneous linear equation on an interval I. The set of solutions is linearly independent on I if $W(y_1, y_2, \ldots, y_n) \neq 0 \ \forall x$ in I

Note that $W(y_1, y_2, \ldots, y_n)$ is either zero or non-zero for all x in I. Thus if it can be shown that for some $x_0 \in I$, $W(y_1, y_2, \ldots, y_n) \neq 0$, then the set of solutions $\{y_1, y_2, \ldots, y_n\}$ is linearly independent on I.

General Solution Of Homogeneous Linear Equation

Let $y_1, y_2, \ldots, y_n$ be a set of fundamental solutions of an nth order homogeneous linear equation on an an interval I. The general solution of the equation on the interval is

$$y = C_1 y_1 + C_2 y_2 + \cdots + C_n y_n$$

where C_i's are arbitrary constants.

Consider a second order homogeneous equation

$$a_2 y'' + a_1 y' + a_0 y = 0 \quad \text{——} ①$$

A solution to equation ① has a property that its second derivative can be expressed as a linear combination of its first and zeroth derivative. One such solution is $y = e^{rx}$, substituting into equation ① we get

$$a_2 r^2 e^{rx} + a_1 r e^{rx} + a_0 e^{rx} = 0$$
$$\Rightarrow e^{rx}(a_2 r^2 + a_1 r + a_0) = 0$$
$$\Rightarrow a_2 r^2 + a_1 r + a_0 = 0 \quad \text{——} ②$$

This means that e^{rx} is a solution to equation ① if r satisfies equation ②, which is known as the auxiliary equation of equation ①.

From equation ②,

$$r_1 = \frac{-a_1 + \sqrt{a_1^2 - 4a_2 a_0}}{2a_2}$$

and $$r_2 = \frac{-a_1 - \sqrt{a_1^2 - 4a_2 a_0}}{2a_2}$$

r_1 or r_2 can be

① distinct real root
② repeated roots
③ complex roots

Distinct Real Roots

If the auxiliary equation has distinct real roots r_1 and r_2, then $y_1 = e^{r_1 x}$ and $y_2 = e^{r_2 x}$ are fundamental solutions to the homogeneous equation, and

$$y = C_1 e^{r_1 x} + C_2 e^{r_2 x},$$

is the general solution

Examples

Solve ① $y'' - 9y = 0$

② $y''' + y'' - 6y' + 4y = 0$

Sol^{n} ①. The auxiliary equation is

$r^2 - 9 = 0$

$\Rightarrow r = \pm 3$

$\therefore$ the general solution is

$$y = C_1 e^{3x} + C_2 e^{-3x}$$

②. Auxiliary equation is $r^3 + r^2 - 6r + 4 = 0$

$\Rightarrow (r-1)(r^2 + 2r - 4) = 0$

$\Rightarrow r = 1$ or $r = -1 + \sqrt{5}$ or $r = -1 - \sqrt{5}$

$\therefore y = C_1 e^{x} + C_2 e^{(-1+\sqrt{5})x} + C_3 e^{(-1-\sqrt{5})x}$

REPEATED ROOTS

If the auxiliary equation has repeated roots, then $y_1 = e^{rx}$ and $y_2 = xe^{rx}$ are fundamental solutions, and $y = c_1e^{rx} + c_2xe^{rx}$ is the general solution of the homogeneous equation

Examples

Solve ① $y'' + 6y' + 9y = 0$

② $y''' + 3y'' - 4y = 0$

Soln ① Auxiliary eqn: $r^2 + 6r + 9 = 0$

$\Rightarrow (r+3)^2 = 0$

$\Rightarrow r = -3$

$\therefore y = c_1e^{-3x} + c_2xe^{-3x}$

②. Auxiliary eqn: $r^3 + 3r^2 - 4 = 0$

$\Rightarrow (r-1)(r^2 + 4r + 4) = 0$

$\Rightarrow (r-1)(r+2)^2 = 0$

$\Rightarrow r = 1$ or $r = -2$

$\therefore y = c_1e^{x} + c_2e^{-2x} + c_3xe^{-2x}$

Complex Roots

If the auxiliary equation has complex roots $r_1 = \alpha + i\beta$ and $r_2 = \alpha - i\beta$, then $y_1 = e^{\alpha x}\cos(\beta x)$ and $y_2 = e^{\alpha x}\sin(\beta x)$ are fundamental solutions, and $y = c_1 e^{\alpha x}\cos(\beta x) + c_2 e^{\alpha x}\sin(\beta x)$ is a general solution to the homogeneous equation.

Examples

Solve ① $y'' - 4y' + 7y = 0$

② $y''' + y'' + 3y' - 5y = 0$

Soln ① Auxiliary eqn: $r^2 - 4r + 7 = 0$

$$\Rightarrow r = \frac{4 \pm \sqrt{16 - 4(7)}}{2}$$

$$\Rightarrow r = \frac{4 \pm \sqrt{-12}}{2}$$

$$\Rightarrow r = 2 + \sqrt{-3} \quad \text{or} \quad r = 2 - \sqrt{-3}$$

$$= 2 + i\sqrt{3} \qquad r = 2 - i\sqrt{3}$$

$$\therefore y = c_1 e^{2x}\cos(\sqrt{3}x) + c_2 e^{2x}\sin(\sqrt{3}x)$$

② Auxiliary eqn: $r^3 + r^2 + 3r - 5 = 0$

$$(r-1)(r^2 + 2r + 5) = 0$$

$$\Rightarrow r = 1 \quad \text{or} \quad r = \frac{-2 \pm \sqrt{4 - 4(5)}}{2}$$

$$= -1 \pm 2i$$

$$\therefore y = c_1 e^{x} + c_2 e^{-x}\cos(2x) + c_3 e^{-x}\sin(2x)$$

NON-HOMOGENEOUS LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

These are equations of the form

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 y' + a_0 y = f(x), \quad f(x) \neq 0$$

METHODS FOR SOLVING NON-HOMOGENEOUS LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

1. UNDETERMINES COEFFICIETS METHOD

We find the complementary solution y_c, which is the solution to the associated homogeneous equation

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 y' + a_0 y = 0.$$

Looking at $f(x)$, we guess the form of the particular solution y_p and solve for undetermined coefficients. The general solution is

$$y = y_c + y_p$$

The method of undetermined coefficients only works for linear equations with constant coefficients, and it can only be used if $f(x)$ has the property that all its derivatives can be written in terms of a finite number of functions. Functions with this property are polynomials, exponentials, sines, cosines, and the products of these functions.

A Guide For Guessing The Form Of y_p

- If $f(x) = Cx^m e^{rx}$, where m is a non-negative integer, we choose

$$y_p = x^s(A_m x^m + \cdots + A_1 x + A_0)e^{rx}$$

with $*\ s=0$ if r is not the root of the associated auxiliary equation

$*\ s=1$ if r is a simple real root of the associated auxiliary equation

$*\ s=2$ if r is a repeated root of the associated auxiliary equation.

- If $f(x) = \begin{cases} Cx^m e^{\alpha x}\cos\beta x \\ Cx^m e^{\alpha x}\sin\beta x \end{cases}$ with $\beta \neq 0$, then

$$y_p = x^s(A_m x^m + \cdots + A_1 x + A_0)e^{\alpha x}\cos(\beta x) + x^s(B_m x^m + \cdots + B_1 x + B_0)e^{\alpha x}\sin(\beta x)$$

with $*\ s=0$ if $\alpha + i\beta$ is not a root to the associated auxiliary equation

$*\ s=1$ if $\alpha + i\beta$ is complex root to the associated auxiliary equation.

Example 1.

Find the particular solution to

1. (a) $y'' + 2y' - y = 10$
   (b) $y'' + y' = 2$
2. (a) $2y' + y = 3x^2$
   (b) $y'' + 4y' - 2y = 2x^3 - 3x + 6$
3. (a) $y'' + y' = 2^x$
   (b) $y'' - 5y' + 4y = 8e^x$
4. (a) $y'' - y' + y = 2\sin(3x)$
   (b) $y''' - 2y'' + y = \cos(x)$
   (c) $y'' + y = \cos(x)$

5. (a) $y''' + y'' - 2y = x e^x$

(b) $y'' + y = x \cos x$

(c) $y'' - 2y' + 5y = e^x \cos(2x)$

(d) $y' - 2y = x \sin(x) e^x$

Example 2

Find the general solution to

$$y'' + 9y = 2\cos(3x)$$

Soln Example 1

1.(a) $y'' + 2y' - y = 10$

N.B $10 = 10e^{0 \cdot x}$ and $r=0$ is not a root to the associated auxiliary equation, thus we choose

$y_p = A$, where A is a constant

$\Rightarrow y_p' = 0$

$\Rightarrow y_p'' = 0$

substituting into the equation, we get

$$-A = 10 \Rightarrow A = -10$$

$\therefore y_p = -10$

(b) $y'' + y' = 2$

since $2 = 2e^{0 \cdot x}$ and $r=0$ is a root to the associated auxiliary equation, we choose

$y_p = xA$

$\Rightarrow y_p' = A$

$\Rightarrow y_p'' = 0$

Substituting into the equation, we get

$A = 2$

$\therefore y_p = 2x$

$2y' + y = 3x^2$

$r = 0$ is not the root of the associated auxiliary eqn, so we choose

$y_p = Ax^2 + Bx + C$

$\Rightarrow y_p' = 2Ax + B$

Substitution into the eqn gives

$4Ax + 2B + Ax^2 + Bx + C = 3x^2$

Equating coefficients of powers of x, we have

$A = 3, \quad 4A + B = 0, \quad 2B + C = 0$

$\Rightarrow B = -12$ and $C = 24$

$\therefore y_p = 3x^2 - 12x + 24$

(b) $y'' + 4y' - 2y = 2x^3 - 3x + 6$

N.B $r = 0$ is not the root of the associated auxiliary equation, so let

$y_p = Ax^3 + Bx^2 + Cx + D$

$\Rightarrow y_p' = 3Ax^2 + 2Bx + C$

$\Rightarrow y_p'' = 6Ax + 2B$

substituting into the equation, we get

$$6Ax+2B+12Ax^2+8Bx+4C-2Ax^3-2Bx^2-2Cx-2D = 2x^3-3x+6$$

Equating coefficients of powers of x, we get

x^3: $-2A = 2 \Rightarrow A = -1$

x^2: $12A-2B=0 \Rightarrow B=-6$

x^1: $6A+8B-2C=-3 \Rightarrow C=-5\frac{1}{2}$

x^0: $2B+4C-2D=6 \Rightarrow D=-60$

$\therefore y_p = -x^3-6x^2-5\frac{1}{2}x-60$

3. (a) $y''+y'=2^x$

N.B $2^x = e^{x\ln 2}$ and $r=\ln 2$ is not the root of the associated auxiliary equation, so we choose

$$y_p = A2^x$$
$$\Rightarrow y_p' = A\ln(2)2^x$$
$$\Rightarrow y_p'' = A(\ln 2)^2 2^x$$

Substituting into the eqⁿ, we get

$$A(\ln 2)^2 2^x + A\ln(2)2^x = 2^x$$

Equating the coefficients of 2^x we get

$$A(\ln(2))^2 + A\ln(2) = 1$$
$$\Rightarrow A = \frac{1}{(\ln 2)^2+(\ln 2)}$$

$$\therefore y_p = \frac{2^x}{(\ln 2)^2+(\ln 2)}$$

(b) $y'' - 5y' + 4y = 8e^x$

N.B $f(x) = 8e^x = 8e^{1\cdot x}$

$r = 1$ is the root of the associated auxiliary eqn, so choose

$$y_p = Axe^x$$
$$\Rightarrow y_p' = Axe^x + Ae^x$$
$$\Rightarrow y_p'' = Axe^x + 2Ae^x$$

Substituting into the equation, we get

$$Axe^x + 2Ae^x - 5Axe^x - 5Ae^x + 4Axe^x = 8e^x$$
$$\Rightarrow -3Ae^x = 8e^x$$
$$-3A = 8 \Rightarrow A = -8/3$$

$$\therefore y_p = -8/3\, xe^x$$

4. (a) $y'' - y' + y = 2\sin(3x)$

N.B $\pm i3$ is not a root to the auxiliary equation, so we choose

$$y_p = A\cos(3x) + B\sin(3x)$$
$$\Rightarrow y_p' = -3A\sin(3x) + 3B\cos(3x)$$
$$\Rightarrow y_p'' = -9A\cos(3x) - 9B\sin(3x)$$

Substituting into the equation, we get

$$-9A\cos(3x) - 9B\sin(3x) + 3A\sin(3x) - 3B\cos(3x) + A\cos 3(x) + B\sin(3x) = 2\sin(3x)$$
$$\Rightarrow -8A\cos(3x) - 3B\cos(3x) - 8B\sin(3x) + 3A\sin(3x) = 2\sin(3x)$$

$\Rightarrow -8A - 3B = 0$ and $-8B + 3A = 2$

$\Rightarrow \begin{cases} -24A - 9B = 0 \\ -64B + 24A = 16 \end{cases} \Rightarrow -73B = 16$

$\Rightarrow B = \frac{-16}{73}$

$\Rightarrow A = \frac{6}{73}$

$\therefore y_p = \frac{6}{73}\cos(3x) - \frac{16}{73}\sin(3x)$

(b) $y''' - 2y'' + y = \cos(x)$

N.B $\pm i$ is not the ~~root~~ to the associated homogeneous equation, so we choose

$y_p = A\cos x + B\sin x$

$\Rightarrow y_p' = -A\sin x + B\cos x$

$\Rightarrow y_p'' = -A\cos x - B\sin x$

$\Rightarrow y_p''' = A\sin x - B\cos x$

Substituting into the equation, we get

$A\sin x - B\cos x + 2A\cos x + 2B\sin x + A\cos x + B\sin x = \cos x$

$\Rightarrow A + 2B + B = 0 \Rightarrow A + 3B = 0 \Rightarrow A = -3B$

and $-B + 2A + A = 1 \Rightarrow -B + 3A = 1$

$\Rightarrow -B - 9B = 1$

$\Rightarrow -10B = 1$

$\Rightarrow B = -\frac{1}{10} \Rightarrow A = \frac{3}{10}$

$\therefore y_p = \frac{3}{10}\cos x - \frac{1}{10}\sin x$

(c) $y'' + y = \cos x$

N.B $\pm i$ is the root of the associated homogeneous equation, so we choose

$$y_p = Ax\cos x + Bx\sin x$$
$$\Rightarrow y_p' = A\cos x - Ax\sin x + B\sin x + Bx\cos x$$
$$\Rightarrow y_p'' = -A\sin x - A\sin x - Ax\cos x + B\cos x + B\cos x - Bx\sin x$$
$$= -2A\sin x - Ax\cos x + 2B\cos x - Bx\sin x$$

Substituting into the equation, we get

$$-2A\sin x - Ax\cos x + 2B\cos x - Bx\sin x + Ax\cos x + Bx\sin x = \cos x$$
$$\Rightarrow -2A\sin x + 2B\cos x = \cos x$$
$$\Rightarrow A = 0 \quad \text{and} \quad B = \tfrac{1}{2}$$

$$\therefore y_p = \tfrac{1}{2}x\sin x$$

5. (a) $y''' + y'' - 2y = xe^x$

Note that $r = 1$ is the root to the associated homogeneous equation, so we choose

$$y_p = (Ax^2 + Bx)e^x$$
$$\Rightarrow y_p' = (Ax^2 + Bx)e^x + (2Ax + B)e^x$$
$$\Rightarrow y_p'' = (Ax^2 + Bx)e^x + 2(2Ax + B)e^x + 2Ae^x$$
$$\Rightarrow y_p''' = (Ax^2 + Bx)e^x + 3(2Ax + B)e^x + 6Ae^x$$

Substituting into the equation, we get

$$(Ax^2 + Bx)e^x + 3(2Ax + B)e^x + 6Ae^x + (Ax^2 + Bx)e^x + 2(2Ax + B)e^x + 2Ae^x - 2(Ax^2 + Bx)e^x = xe^x$$

$\Rightarrow 5(2Ax+B)e^x + 8Ae^x = xe^x$

$\Rightarrow 10A = 1$ and $5B + 8A = 0$

$\Rightarrow A = \frac{1}{10}$ and $B = \frac{-4}{25}$

$\therefore y_p = \left(\frac{x^2}{10} - \frac{4x}{25}\right)e^x$

(b) $y'' + y = x\cos x$

N.B $r = \pm i$ is the root of the associated auxiliary equation, so we choose

$y_p = (Ax^2+Bx)\cos x + (Cx^2+Dx)\sin x$

$\Rightarrow y_p' = -(Ax^2+Bx)\sin x + (2Ax+B)\cos x + (Cx^2+Dx)\cos x + (2Cx+D)\sin x$

$\Rightarrow y_p'' = -(Ax^2+Bx)\cos x - 2(2Ax+B)\sin x + 2A\cos x - (Cx^2+Dx)\sin x + 2(2Cx+D)\cos x + 2C\sin x$

Substituting into the equation, we get

$-(Ax^2+Bx)\cos x - 2(2Ax+B)\sin x + 2A\cos x - (Cx^2+Dx)\sin x + 2(2Cx+D)\cos x + 2C\sin x + (Ax^2+Bx)\cos x + (Cx^2+Dx)\sin x = x\cos x$

$\Rightarrow -2(2Ax+B)\sin x + 2A\cos x + 2(2Cx+D)\cos x + 2C\sin x = x\cos x$

$-4A = 0 \Rightarrow A = 0$

$-2B + 2C = 0 \Rightarrow B = C$

$2A + 2D = 0 \Rightarrow D = 0$

$4C = 1 \Rightarrow C = \frac{1}{4} \Rightarrow B = \frac{1}{4}$

$\therefore y_p = \frac{1}{4}x\cos x + \frac{1}{4}x^2\sin x$

(C) $y'' - 2y' + 5y = e^x \cos(2x)$

N.B $1 \pm 2i$ is the root to the associated auxiliary equation, we choose

$$y_p = Axe^x \cos 2x + Bxe^x \sin 2x$$

$$\Rightarrow y_p' = Axe^x \cos 2x + Ae^x \cos 2x - 2Axe^x \sin 2x + Bxe^x \sin 2x + Be^x \sin(2x) + 2Bxe^x \cos(2x)$$

$$\Rightarrow y_p'' = Axe^x \cos(2x) + 2Ae^x \cos(2x) - 4Axe^x \sin(2x) - 4Ae^x \sin(2x) - 4Axe^x \cos 2x + Bxe^x \sin 2x + 2Be^x \sin(2x) + 4Bxe^x \cos(2x) + 4Be^x \cos(2x) - 4Bxe^x \sin(2x)$$

$$= -3Axe^x \cos 2x + 2Ae^x \cos(2x) - 4Axe^x \sin(2x) - 4Ae^x \sin 2x - 3Bxe^x \sin(2x) + 2Be^x \sin(2x) + 4Bxe^x \cos(2x) + 4Be^x \cos(2x)$$

Substituting into the equation we get

$$-3Axe^x \cos(2x) + 2Ae^x \cos(2x) - 4Axe^x \sin(2x) - 4Ae^x \sin(2x) - 3Bxe^x \sin(2x) + 2Be^x \sin(2x) + 4Bxe^x \cos(2x) + 4Be^x \cos(2x) - 2Axe^x \cos 2x - 2Ae^x \cos(2x) + 4Axe^x \sin 2x - 2Bxe^x \sin(2x) - 2Be^x \sin(2x) - 4Bxe^x \cos(2x) + 5Axe^x \cos(2x) + 5Bxe^x \sin(2x) = e^x \cos(2x)$$

$$\Rightarrow -4Ae^x \sin(2x) + 4Be^x \cos(2x) = e^x \cos(2x)$$

$$\Rightarrow -4A = 0 \Rightarrow A = 0$$

and $4B = 1 \Rightarrow B = \frac{1}{4}$

$$\therefore y_p = \frac{1}{4} x e^x \sin(2x)$$

d) $y' - 2y = x\sin(x)e^x$

N.B $\alpha \pm i$ is not a root of the associated auxiliary equation, so we choose

$$y_p = (Ax+B)e^x\cos x + (Cx+D)e^x\sin x$$
$$\Rightarrow y_p' = (Ax+B)e^x\cos x - (Ax+B)e^x\sin x + Ae^x\cos x + (Cx+D)e^x\sin x + (Cx+D)e^x\cos x + Ce^x\sin x$$

Substituting into the equation, we get

$$(Ax+B)e^x\cos x - (Ax+B)e^x\sin x + Ae^x\cos x + (Cx+D)e^x\sin x + (Cx+D)e^x\cos x + Ce^x\sin x - 2(Ax+B)e^x\cos x - 2(Cx+D)e^x\sin x = xe^x\sin x$$

$$\Rightarrow -(Ax+B)e^x\cos x - (Ax+B)e^x\sin x + Ae^x\cos x - (Cx+D)e^x\sin x + (Cx+D)e^x\cos x + Ce^x\sin x = xe^x\sin x$$

$$-A + C = 0 \Rightarrow C = A$$
$$-B + A + D = 0 \Rightarrow D - B = \tfrac{1}{2} \Rightarrow B - D = -\tfrac{1}{2}$$
$$-A - C = 1 \Rightarrow -2A = 1 \Rightarrow A = C = -\tfrac{1}{2}$$
$$-B - D + C = 0 \Rightarrow B + D = -\tfrac{1}{2}$$

$$B - D = -\tfrac{1}{2} \quad \& \quad B + D = -\tfrac{1}{2}$$
$$\Rightarrow 2B = -1 \Rightarrow B = -\tfrac{1}{2} \Rightarrow D = 0$$

$$\therefore y_p = \left(-\tfrac{1}{2}x - \tfrac{1}{2}\right)e^x\cos x - \tfrac{1}{2}xe^x\sin x$$

Example 2

$$y'' + 9y = 2\cos(3x)$$

The auxiliary eqn: $r^2 + 9 = 0$

$$\Rightarrow r = \pm 3i$$

$$\Rightarrow y_c = C_1 \cos(3x) + C_2 \sin(3x)$$

We choose $y_p = Ax\cos 3x + Bx\sin 3x$

$$\Rightarrow y_p' = -3Ax\sin 3x + A\cos 3x + 3Bx\cos 3x + B\sin 3x$$

$$\Rightarrow y_p'' = -9Ax\cos 3x - 6A\sin 3x - 9Bx\sin 3x + 6B\cos(3x)$$

Substituting into the equation we get

$$-9Ax\cos 3x - 6A\sin 3x - 9Bx\sin 3x + 6B\cos(3x) + 9Ax\cos 3x + 9Bx\sin 3x = 2\cos(3x)$$

$$\Rightarrow -6A\sin(3x) + 6B\cos(3x) = 2\cos(3x)$$

$$-6A = 0 \Rightarrow A = 0$$

$$6B = 2 \Rightarrow B = \tfrac{1}{3}$$

$$\Rightarrow y_p = \tfrac{1}{3}x\sin(3x)$$

$$\therefore y = C_1\cos(3x) + C_2\sin(3x) + \tfrac{1}{3}x\sin(3x)$$

Superposition Principle and Undetermined Coefficients

Let y_1 be a solution to

$$a y'' + b y' + c y = f_1(x)$$

and y_2 be a solution to

$$a y'' + b y' + c y = f_2(x)$$

then $k_1 y_1 + k_2 y_2$ is a solution to

$$a y'' + b y' + c y = k_1 f_1(x) + k_2 f_2(x)$$, where k_1 and k_2 are constants

Examples

Find the particular solution to

(a) $y'' + 2y' + 4y = x + \cos(2x)$

(b) $y'' + 5y' + 6y = 6x^2 + 10x + 2 + 12e^x$

So let

(a) $y'' + 2y' + 4y = x + \cos(2x)$

Note that $r = 0$ and $r = \pm 2i$ are not roots to the associated auxiliary equation, so we choose

$$y_p = Ax + B + C\cos(2x) + D\sin(2x)$$

$$\Rightarrow y_p' = A - 2C\sin 2x + 2D\cos(2x)$$

$$\Rightarrow y_p'' = -4C\cos(2x) - 4D\sin(2x)$$

Substituting into the equation, we get

$$-4C\cos(2x) - 4D\sin(2x) + 2A - 4C\sin 2x + 4D\cos(2x) + 4Ax + 4B + 4C\cos(2x) + 4D\sin(2x) = x + \cos(2x)$$

$$\Rightarrow 2A - 4C\sin 2x + 4D\cos(2x) + 4Ax + 4B = x + \cos(2x)$$

$\Rightarrow 2A + 4B = 0$

$C = 0$

$D = 1/4$

$A = 1/4 \quad \Rightarrow B = -1/8$

$\therefore y_p = \frac{1}{4}x - \frac{1}{8} + \frac{1}{4}\sin(2x)$

(b) $y'' + 5y' + 6y = 6x^2 + 10x + 2 + 12e^x$

Note that $r = 0$ and $r = 1$ are not roots of the associated auxiliary equation, so let

$y_p = Ax^2 + Bx + C + De^x$

$\Rightarrow y_p' = 2Ax + B + De^x$

$\Rightarrow y_p'' = 2A + De^x$

Substituting into the equation, we get

$2A + De^x + 10Ax + 5B + 5De^x + 6Ax^2 + 6Bx + 6C + 6De^x$

$= 6x^2 + 10x + 2 + 12e^x$

equating coefficients, we get

$2A + 5B + 6C = 2$

$12D = 12 \Rightarrow D = 1$

$10A + 6B = 10$

$6A = 6 \Rightarrow A = 1$

$\Rightarrow B = 0$

$2 + 6C = 2 \Rightarrow C = 0$

$\therefore y_p = x^2 + e^x$